

Backed By Fans

Creator-owned memberships

Working draft · September 2026



Backed By Fans lets creators offer memberships on their own terms. Fans pay for membership time and receive a transferable NFT that carries the membership and its accumulated reward weight. Creators decide what belonging means, from access to a private community to recognition of support for their work.

A membership is a prepaid subscription. Fans can renew manually or, on eligible tiers, authorize periodic refills from their own wallet. Each payment creates a membership or adds time to a selected live membership. As that time is used, the payment is earned by the creator and, where enabled, shared with members and a referrer. A portion goes to the protocol, exclusively to buy and burn the protocol token.

This connects payment to an ongoing membership. Paying for several months does not make the entire payment immediately available to the creator or reward recipients. It funds the months ahead, with value earned throughout that period.

Membership on the creator's terms

A creator can offer one membership or several tiers. Each tier has its own price, payment currency, period length, and reward terms. A musician might offer a supporters' community, a writer might provide access to a private website, and an artist might use membership simply to recognize the people supporting their practice.

The membership NFT gives creators a common way to recognize fans across these experiences. A Discord server, website, or other application can check that a wallet owns a particular membership and that its time has not expired. Creators can build these experiences themselves or use services that support the membership contracts.

The benefits come from the creator. The protocol records membership and handles payments and rewards; it does not deliver content, operate a community, or guarantee the benefits described by a creator.

Terms that stay fixed

When a creator publishes a tier, its core terms become permanent:

- The tier's name and symbol.
- Its payment currency, price per period, and period length.
- Its minimum positive payment.
- The percentages allocated to member rewards, referrals, and the protocol.
- The rules determining how much reward weight each payment creates.

Fans can review these terms before joining. A creator cannot later raise the price of that same tier, change its currency, or reduce its member-reward percentage. Different economic terms require a new tier.

The protocol allocation must be at least 1% of each payment. The creator chooses the tier's allocation within the contract's limits. Member and referral rewards are optional; either percentage can be zero. The creator receives what remains after the applicable allocations.

What can evolve

Creators can update a tier's description, website, and NFT artwork. These updates can affect existing memberships as well as new ones. Artwork can use the built-in styles or a custom onchain design.

Creators can also change membership capacity and the amount of time fans may prepay. Capacity counts membership positions, so two NFTs owned by one wallet occupy two places. A capacity reduction cannot go below the number of occupied places, and a new prepayment limit does not remove time already purchased.

A creator may pause new membership time additions, give complimentary time, revoke unused complimentary time, or lower the creator share on cancellation. Pausing does not stop the membership clock or undo earnings. Live transfers, approvals, maintenance, and claims remain available while paused. Revoking complimentary time preserves any remaining paid time; if no time remains, the position is retired.

Tier ownership can be transferred to a new owner. The new owner takes over these responsibilities and controls while the tier's permanent terms remain in place.

Joining, renewing, and returning

Joining creates a new membership NFT with its own token ID. A wallet can own several independent memberships in the same tier, each with its own time, reward weight, earnings, and referral choice. Creating another membership always creates another ID; it does not merge with an existing position.

Fans choose manual renewal or explicitly authorize a remaining-time target for periodic refill. Renewal selects an existing token ID and extends it before its membership expires. New paid time follows its remaining paid time, and paid time is used before complimentary time. At or after expiration, that ID cannot be renewed. Returning requires a new membership, subject to the tier's capacity and terms.

A tier can also accept contributions instead of setting a fixed price. In that model, creating or renewing a selected membership adds one period. A fan can contribute zero or choose a positive amount that meets the tier's minimum. A zero contribution adds time but funds no rewards or creator earnings and issues no reward weight.

Fixed-price gifts can create a new position for another wallet or fund a renewal of a selected live position. The membership receives the time and reward weight, regardless of who paid. Creators can similarly create a complimentary membership or add grant time to a selected live position. Complimentary time has no payment allocation or new reward weight. Gifts and grants must distinguish creation from extension.

Optional periodic refill

A creator can enable periodic refill when publishing a positive fixed-price tier or later. Enablement is permanent, but it enrolls nobody and moves no funds. Contribution-priced and free tiers do not support it. Only the current NFT owner can enroll a live position, change its target or stop it; an approved NFT operator cannot give this spending consent.

The owner chooses a positive target for total remaining time, including paid and complimentary time. A separate payment-token allowance authorizes the tier to collect funds from that owner's wallet. Anyone can submit a refill transaction while the membership remains live. Someone must submit and pay for that transaction: a funded wallet and allowance alone do not schedule execution or guarantee continuity.

Refill buys whole periods toward the target. With a thirty-day period, a seven-day target and six days remaining buys one period, leaving 36 days. A 45-day target with ten days remaining buys two periods, leaving 70 days. At or above the target, nothing is due. The purchase is bounded by the caller's period limit, available balance, tier allowance, paid-time capacity and numeric limits. If only one whole period is affordable, only one is purchased; less than one affordable period produces no charge. The new payment uses the ordinary allocation, referral and reward-curve rules.

Coverage estimates are shared, not reserved. At a price of 20, a balance of 55 covers 2.75 equivalent periods, while an allowance of 40 permits at most two whole periods. Several positions can show the same available funds; execution rechecks what remains. A pending referral choice locks on the first successful paid purchase. If a manual payment locks a choice first, refill uses that locked choice.

Stopping refill preserves existing time and weight. Revoking allowance blocks collection but leaves enrollment configured; restoring allowance can resume collection for a still-live position. Pausing likewise blocks payments while allowing target changes, stopping, allowance revocation and cancellation. Transfers, cancellation, retirement and successful removal of nonzero complimentary time clear enrollment. After a transfer or grant revocation, the current holder must explicitly enroll again. No stored reason is needed for the ordinary "Periodic refill off" state; transaction history records the action.

Refill must complete strictly before expiration. At expiration, even a funded, approved and enrolled position cannot buy more time or recover its former identity. Mandatory accounting and claims never collect refill payments. A failed payment rolls back its own attempted accounting catch-up; separate maintenance can still save progress for other positions.

Paid-time limits

Let N be the tier's maximum prepaid periods and D its period duration. For a finite limit, every paid addition must leave remaining paid time strictly below $(N + 1) \times D$. This rule applies to manual creation and renewal, sponsorships, contribution payments and periodic refill; complimentary time does not consume paid capacity. $N = 0$ means unlimited, subject to ordinary numeric and expiration bounds.

For $N = 1$ and a thirty-day period, 29 remaining paid days plus a new thirty-day period gives 59 days and is allowed. Exactly 60 days is rejected. The same exclusive limit applies for every positive N . Newly configured refill targets may be no greater than $N \times D$, although buying whole periods can leave a fractional-period overhang above that target. Lowering the cap never erases purchased time or an existing enrollment target: it blocks additions until a whole period fits. Raising the cap can make that unchanged target refillable again.

Moving a live membership

A transfer moves the entire membership position to its new owner. Its token ID keeps the remaining paid and complimentary time, accumulated weight, reward eligibility, unclaimed member rewards, funding history, and locked referral choice. The recipient can already own other memberships in that tier; the transferred position remains separate.

The new owner gains membership access and the authority to claim its rewards and renew it. The former owner loses those rights. Original payment records remain historical evidence of who funded the position; they do not give a former owner or payer authority over it.

An owner can approve another address to transfer an NFT, or approve an operator for their NFTs. Those approvals authorize transfers and member cancellation. They do not authorize owner-only renewal, refill enrollment, reward claims, or creator actions. A token-specific approval and refill enrollment clear when the NFT transfers, including a self-transfer. A person may still sponsor a renewal under the same rules available to any payer.

Live transfers do not settle rewards or run accounting maintenance. A backlog of funding or expiration checkpoints cannot block an otherwise valid transfer, including while the tier is paused. The transfer leaves the position's stored accounting and schedules unchanged and checks expiration directly against the transaction's timestamp.

Expiration and permanent retirement

Access ends at the membership's expiration timestamp. From that timestamp onward, the NFT cannot transfer or receive more time, even if it is still present in the wallet awaiting maintenance. Applications must verify current ownership and active membership status; NFT possession alone is insufficient.

Anyone can process maintenance. It settles funding chronologically through each expiration before removing that position's weight. A late maintenance transaction therefore gives the membership the rewards it earned through its actual expiration, without awarding it rewards for time after expiration. Funding ending at the same timestamp, including its rounding remainder, is settled before any position expiring then is retired.

Retirement permanently burns the NFT, destroys its reward weight, clears its live membership and referral association, and releases its occupied place once. Its already-earned member credit moves, without rounding away fractions, to a separate balance belonging to the owner at expiration. Funding history remains available under the retired ID.

Returning starts a separate position with a new ID and fresh referral state. A positive payment earns new weight at the tier's current reward curve. Retirement and refunds never reduce lifetime payment volume or reopen an earlier point on that curve. A new membership does not inherit the retired position's weight or consume its owner's retired reward balance.

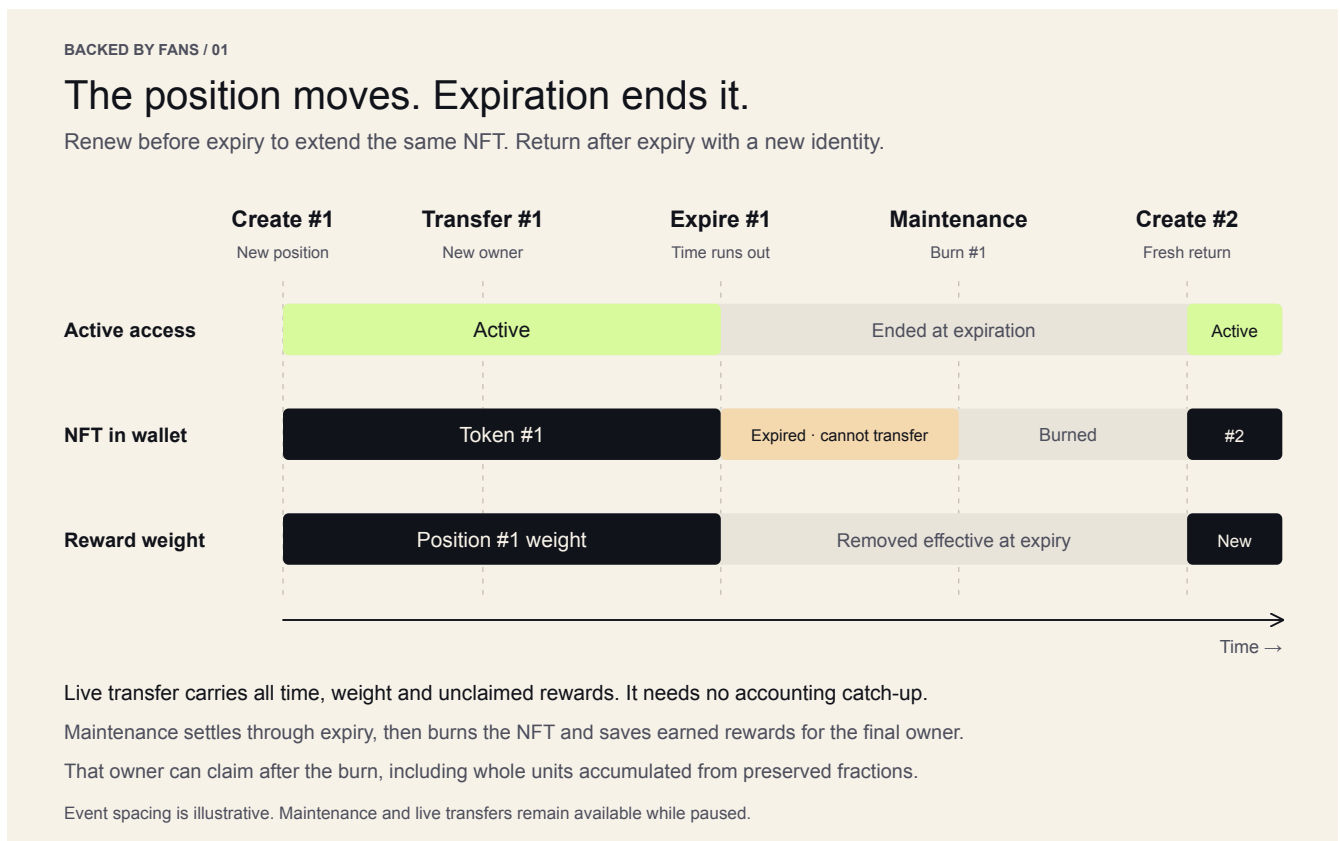


Figure 1. Access ends at expiration. Maintenance completes permanent retirement at that historical boundary; returning creates a new position.

How payments become earnings

Each payment has four possible destinations:

Recipient	What the allocation supports
Creator	Earnings from the membership
Members	Optional rewards shared among eligible memberships
Referrer	Optional rewards for the membership's recorded referrer
Protocol	Fees available for protocol-token buybacks and burns

The amounts are allocated when the payment is made. They are earned as its membership time is used.

This distinction matters when a fan prepays. A payment for an additional month is reserved for that month. It does not become immediately earned simply because the fan paid early. The same timing applies to creator earnings, member rewards, referral rewards, and protocol fees.

Once earned, creator proceeds and rewards can be claimed. Accrual describes how the amount grows over time; it does not mean tokens are transferred to every recipient's wallet every second. Claims and protocol-fee release happen through transactions.

All four allocations use the tier's payment currency. A membership priced in one currency does not automatically pay its member rewards in the protocol token.

A payment over thirty days

Consider a tier charging 100 units of its payment currency for thirty days. Its terms allocate 70% to the creator, 20% to members, 5% to a referrer, and 5% to the protocol. These are illustrative percentages, not protocol defaults.

Assume the fan has no existing paid time and has a recorded referrer. With no refund or other change, the payment is earned as follows:

Time used	Creator	Member rewards	Referrer	Protocol	Unearned
At payment	0	0	0	0	100
Fifteen days	35	10	2.5	2.5	50
Thirty days	70	20	5	5	0

The member-reward column is the amount shared by the pool, not an amount promised to any one fan. Each eligible membership receives a portion according to its reward weight during the time those rewards are earned.

If the membership has no recorded referrer, the referral portion goes to the creator. In this example, the allocation would become 75% creator, 20% members, and 5% protocol.

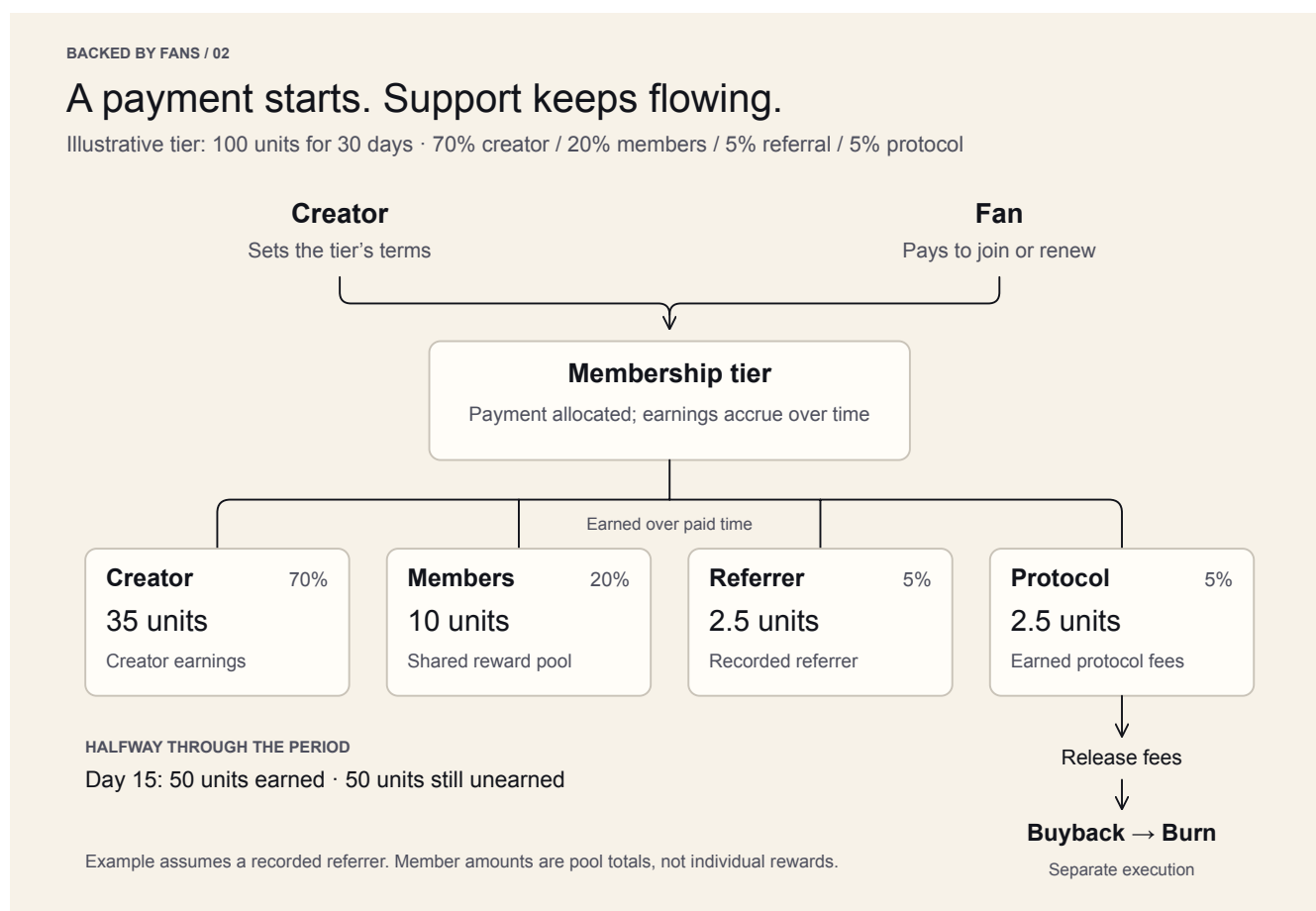


Figure 2. All four allocations accrue over paid time. Earned protocol fees are released for separate buyback and burn execution.

Member cancellation

The current NFT owner or its currently approved token or operator-wide delegate can cancel a live membership, including while the tier is paused. Creator authority alone does not authorize cancellation. The payout always goes to the current NFT owner, including when another wallet originally paid or gifted the membership.

At tier creation, the creator explicitly selects a retention percentage between zero and 100%. The creator can only lower this percentage later, including after a transfer of creator authority. Lower terms immediately apply to existing positions. For unused gross funding G and retention B in basis points, the owner receives $\text{floor}(G \times (10,000 - B) / 10,000)$; the creator earns the remainder. At zero retention the owner receives all unused gross funding, and at 100% the owner can still exit with no payout. Retained proceeds enter ordinary withdrawable creator earnings without another allocation split.

In the thirty-day example, cancellation after fifteen days leaves approximately 50 units of unused funding. At 30% retention, approximately 35 units return to the owner and 15 become creator earnings. Exact amounts follow the currency's smallest units and rounding rules. Already earned creator proceeds, rewards and protocol fees are preserved; fractional cancellation reserves remain reserved and are not paid out twice. Lifetime gross never decreases.

Cancellation clears all paid and complimentary time, burns the NFT, destroys its weight and releases its occupied place exactly once. Earned member credit, including fractions, remains claimable by the final owner. It never purchases more time. Grant revocation is separate: the creator can revoke unconsumed complimentary time, preserving paid time, but a successful nonzero revocation stops periodic refill and requires the holder to enroll again.

The native cancellation action protects the reviewed owner, a minimum owner refund and a deadline no later than the instant before expiration. Its bounded preview reports the current estimate separately from a conservative minimum at that deadline, including scheduled rate changes. Incomplete previews require accounting maintenance before review. If terms, ownership or timing violate the submitted bounds, the entire cancellation reverts. The ERC-5643 adapter uses the same owner/operator cancellation rules with a zero minimum and its fixed accounting budget; it is not the protected native quote flow.

Sharing membership rewards

Creators can dedicate part of each payment to their members. This allows the community to participate in the membership activity it helps sustain.

Rewards can come from new fans joining, existing fans renewing, and gifts. A payment funding a position can also contribute to that position's rewards; its current owner need not have made the payment. New reward weight participates in rewards earned after that weight is created, including later earnings from membership periods already underway. It does not receive a share of rewards earned before it existed.

Reward weight

Positive payments add reward weight to the membership being funded. The contracts call this weight "shares." Shares determine how the member-reward pool is divided and move with the NFT as part of the entire position. They cannot be transferred separately and do not represent ownership of the creator's business.

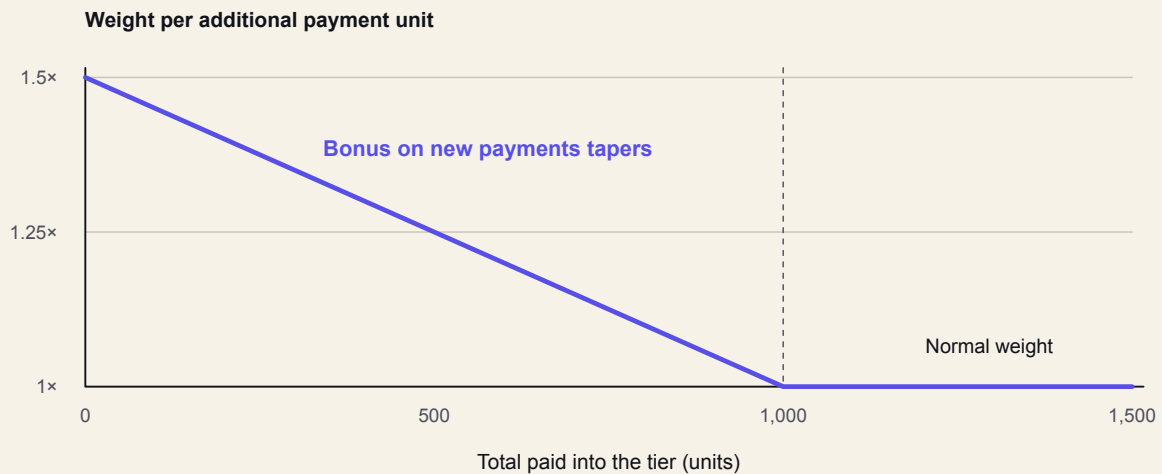
For example, if two eligible memberships have equal weight throughout an interval, they split the rewards earned during that interval equally. If one has twice the weight of the other, it receives two-thirds and the other receives one-third. New payments and changes in eligibility can change those proportions for later earnings.

Creators can choose to give early support additional weight. Early payments receive more reward weight per unit paid. This bonus decreases as total payments into the tier grow. Advancing along the curve does not reduce weight already held by a live position. The starting bonus and payment threshold are fixed when the tier is created.

The benefit follows the amount paid into the tier, not a calendar deadline or a fixed number of people. A payment that spans different points on the curve receives the corresponding weight across those points. Renewals can add weight to a live position, and new issuance depends on where the tier is on its curve when the payment succeeds. Transferring a membership issues no additional weight.

Earlier support can receive more weight.

Illustrative curve: 1.5× starting weight, reaching 1× after 1,000 units paid into the tier.



First 100 units paid

147.5 shares added

Includes the average bonus across that payment.

100 units paid after the threshold

100 shares added

Live positions keep their existing weight until retirement.

Live transfers preserve weight. Retirement removes it; a new position starts on the current curve.

Shares determine the split of eligible member rewards; they are not a promised reward amount.

Figure 3. The bonus applies to new payments. This example uses a 1.5× starting boost and a 1,000-unit threshold; creators choose their tier's terms.

Retirement and earned rewards

An expired membership stops earning at its expiration boundary. Delayed maintenance may leave old weight visible in unsettled accounting, but cannot extend its earning period. Maintenance applies each historical boundary before advancing to later earnings.

Retirement destroys the position's weight permanently while preserving all member rewards already earned. Those rewards belong to the owner at expiration, including earnings accumulated before a transfer. A refund retires the position at the cancellation timestamp with the same protection for earned credit.

Retired rewards are held separately for each wallet and tier, in that tier's payment currency. The wallet can claim them without owning an NFT. Credits from several retired positions are added before rounding to a whole smallest payment unit; any remaining fraction is retained for later claims. Creating a new membership neither resets that balance nor adds it to the new position's weight.

Withdrawal of already-settled retired credit does not require global accounting catch-up. It can proceed while the tier is paused or other positions still await maintenance.

If rewards accrue while no membership has eligible weight, those amounts remain reserved. They do not become additional creator earnings.

Member rewards depend on actual payments, the tier's chosen percentage, and the eligible weight sharing the pool. No particular reward amount is guaranteed.

Referral rewards

A creator can also dedicate a portion of membership payments to referrals.

A membership records its referral choice on its first qualifying positive payment by its owner. That choice can be a referrer or no referrer and stays locked to that token ID. It remains unchanged through renewal and transfer, including a transfer to the recorded referrer. A fresh position has its own referral choice.

Newly locking a referral to the paying owner is rejected. Transferring a membership to its already-locked referrer does not change the referral or prevent renewal.

A gift creating a new membership leaves its referral choice unset, even if the recipient owns another membership with a locked referrer. A gift renewing an existing position follows that position's recorded choice without replacing it. Zero contributions and complimentary grants do not lock a referral. When no referrer applies, the creator receives the unused referral portion.

Referral rewards follow the same time-based model as other earnings. A referrer earns their allocation while the referred membership's funded time is used and can claim the earned amount in that tier's payment currency.

Fixed tier contracts

A single factory creates each tier as a standard ERC-1167 minimal proxy pointing to one fixed implementation. Every tier keeps its own balances, memberships, creator authority, and economic settings. Initialization happens once, atomically with creation. The shared implementation cannot be initialized, and neither the proxy target nor the implementation is upgradeable. Sharing code reduces deployment work without allowing the factory owner to rewrite a tier's fixed terms.

Managing several memberships

Membership lists identify each position by its chain, tier, and token ID. The application chooses a page size and keeps all pages of a snapshot at the same block. Transfers and burns can change list order, so a refresh starts ownership discovery again. An incomplete list is labeled as partial; its balances do not imply a complete wallet total.

Claim all discovers the wallet's memberships and tier balances, then simulates and estimates transactions to choose batches that fit the network. Larger portfolios can require additional wallet confirmations. A rejected transaction leaves confirmed batches completed, and pressing Claim all again claims the remaining rewards. Execution verifies ownership again; the application rechecks captured positions after maintenance and transfers without adding newly received memberships to an ongoing claim. Retired, referral, and creator balances are counted once per tier in each transaction, including for wallets with no remaining NFTs.

Callers choose the accounting work budget for each maintenance or membership transaction. Custom methods impose no fixed step, selection, or page maximum. Standard ERC-5643 renewal and cancellation instead attempt up to 25 checkpoints; larger backlogs require standalone maintenance, and a failed adapter call rolls back all work. Each funding start, funding end, or membership retirement counts as one step. Calls save completed progress, so anyone can continue a large backlog through repeated transactions, including while paused. "Complete" means caught up through that transaction's timestamp; newly elapsed time can create more work.

Purchases, refills, renewals, gifts, contributions, grants, revocations, and cancellations all maintain the expiration schedule. Changes to time, weight, or funding require accounting catch-up first. Complimentary and zero-contribution memberships also have expiration entries. If an atomic operation cannot finish its required catch-up, it reverts; explicit maintenance is the route that saves partial progress. Transfers, approvals, refill configuration and already-settled retired withdrawals do not have this prerequisite.

Previews report their timestamp, progress, and completeness using a caller-selected work budget. A complete preview does not guarantee that a later transaction has enough gas or work budget. Unavailable or incomplete reads must not be presented as zero rewards or proof that a wallet has no membership.

The protocol token

The protocol token has a supporting role in Backed By Fans. Memberships provide the reason to participate: supporting a creator, belonging to a community, and receiving the benefits that creator offers.

Earned protocol fees connect that membership activity to the token. Fees are used to buy the protocol token and burn the purchased amount, reducing its supply. Protocol fees accrue with membership time; buybacks and burns happen in separate transactions.

The protocol token will be made available as a membership payment currency. Creators can then publish tiers priced in that token, with creator earnings and member rewards paid in the same currency. Protocol fees received in the protocol token will be burned directly.

Responsibilities and limits

The contracts enforce each tier's fixed economic terms, membership time, reward rules, and payment accounting. Creators remain responsible for their membership benefits and for the controls they retain, including changes to descriptions and artwork, grants, revocations, and reductions to their cancellation share. Expiration maintenance is permissionless and does not depend on the creator remaining available.

The protocol also has administrative controls. Its authority manages which payment currencies can be used for new tiers, their minimum positive payments, and buyback settings and pauses. At launch, a trusted operator selects buyback routes and execution terms off-chain and submits them with each purchase. The authority can later enable public execution under configured routes, price bounds, batch limits, and cooldowns. Public policies may remain valid indefinitely; expiry and total spending budgets are optional. These controls do not give creators a way to rewrite the fixed economic terms of an existing tier.

The fee vault remains permanent and always burns the protocol tokens acquired by a successful buyback. A separate strategy module controls execution, prices, routes and caller authorization. The vault checks balances and the burn; it does not guarantee a fair price. Selecting a module therefore grants it economic authority over the buyback funds made available to it.

The protocol Safe can replace that module after a 48-hour proposal delay. Activation requires paused buybacks and leaves them paused while the new module is configured and reviewed. The Safe can also propose permanently disabling replacement, with a seven-day review period and cancellation available before finalization. Finalizing is irreversible: a defective frozen module may strand buyback funds. Global pause remains available, and the module's own settings and authorities can still change.

In the initial Pons module, operator minimum outputs are absolute. Public execution uses minimum exchange rates applied to the amount actually spent, including partial fills. Returned input remains available for a later buyback and does not consume a finite policy budget.

Using the protocol also depends on the underlying blockchain and payment currencies. Transactions can require gas and take time to complete. Payment tokens may have restrictions imposed by their own issuers. Smart-contract bugs, unavailable services, and problems with third-party integrations can affect the experience. Buybacks additionally depend on market liquidity and execution conditions.

Balances that depend on newly earned rewards may need accounting catch-up before a claim. Already-settled retired balances remain independently claimable. Integrations must distinguish stored accounting, a complete projection, and current ownership.

Contract references

This draft describes the transferable membership lifecycle, member cancellation and periodic refill in the linked contract sources and the transfer (specs/005-transferable-memberships/spec.md) and cancellation/refill (specs/006-member-cancellation-refill/spec.md) specifications. It does not establish the deployment or launch status of a particular network.

- Membership terms, time, NFTs, and claims (contracts/src/MembershipTier.sol)
- Tier creation and payment-currency configuration (contracts/src/MembershipFactory.sol)
- Time-based earnings and reward accounting (contracts/src/libraries/VestingLedger.sol)
- Independent expiration scheduling (contracts/src/libraries/ExpirationSchedule.sol)
- Early-support reward weight (contracts/src/libraries/RewardCurve.sol)
- Protocol-fee custody, module governance, and burning (contracts/src/ProtocolBuybackVault.sol)
- Initial Pons buyback strategy and price policy (contracts/src/PonsBuybackModule.sol)
- Accounting, fee release, and buyback execution (contracts/src/ProtocolBurnRouter.sol)

Public payment reporting

Each tier exposes a bounded, read-only preview of lifetime payments received, cumulative refunds and payouts, and funds still accruing or already earned. The preview follows the same chronological funding and expiration boundaries as maintenance and reports whether its requested time has been reached. Member earnings include preserved rewards from retired memberships; unassigned funding and rounding reserves are reported separately from beneficiary balances. Protocol payouts represent fees released to the buyback vault, not completed trades or burns. The application groups these figures by payment currency and distinguishes future accrual, earned-but-unclaimed balances, and actual payouts. Available buyback funds include unspent fees already in the vault and earned protocol fees awaiting release. A buyback transaction can release those earned fees before spending them; unearned funding remains reserved and is not available for buybacks.

Uncommitted source snapshot SHA-256

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